

# A scalar-matrix counterexample to Conjecture 00000002343

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## Abstract

For every random pencil of one-by-one matrices, the number of eigenvalue collisions is identically zero, independently of the probability law. Its expectation therefore differs from the conjectured value  $\pi/4$ . The supplied Lean project verifies the actual characteristic polynomial, absence of repeated roots, collision count, and expectation.

## 1 Statement and scope

The source conjecture asserts that the expected number of coincidences (crossings) of eigenvalue curves of a random matrix pencil  $A + \lambda B$  is  $(\pi/4)n$ . It imposes no lower dimension restriction beyond being a matrix pencil, no asymptotic qualification, and no probability law. We refute this asserted equality at the admissible dimension  $n = 1$ . The counterexample works for every law of  $A$  and  $B$ , including nondegenerate scalar Gaussian laws; it does not rely on choosing deterministic random matrices.

At a parameter  $\lambda$ , a collision of distinct eigenvalue branches requires a repeated eigenvalue. For a matrix  $M$ , this is a common zero of its characteristic polynomial  $p_M(z) = \det(zI - M)$  and its derivative. We show that no such zero exists at any parameter. Consequently, whether one counts collision parameters, colliding pairs, or transverse crossings, the count in this example is zero.

## 2 Counterexample

**Theorem 1.** *Let  $(\Omega, \mathcal{F}, \mu)$  be any probability space and let  $a, b : \Omega \rightarrow \mathbb{C}$  be any scalar random variables. For the one-by-one pencil*

$$M_\omega(\lambda) = [a(\omega) + \lambda b(\omega)], \quad \lambda \in \mathbb{R},$$

*the collision set is empty for every  $\omega$ . The expected collision count is 0, not  $\pi/4$ .*

*Proof.* Fix  $\omega$  and  $\lambda$ . Direct computation of the one-by-one determinant gives

$$p_{M_\omega(\lambda)}(z) = z - a(\omega) - \lambda b(\omega), \quad p'_{M_\omega(\lambda)}(z) = 1.$$

Thus the unique eigenvalue is  $a(\omega) + \lambda b(\omega)$ , and it is simple. Since the derivative is never zero, no common zero of  $p$  and  $p'$  exists. The set of real parameters with a collision is therefore empty. Its cardinality  $N(\omega)$  is zero, and the random variable  $N$  is the measurable constant zero. Hence

$$\mathbb{E}_\mu[N] = \int_\Omega 0 \, d\mu = 0.$$

The conjecture instead gives  $(\pi/4) \cdot 1 > 0$ , a contradiction. □

### 3 Formal verification

The file `lean/Main.lean` uses Mathlib’s actual `Matrix.charpoly` and the Bochner integral. It proves the following chain, without a hypothesis asserting the desired spectral conclusion:

- `pencil_charpoly` evaluates the characteristic polynomial of arbitrary complex one-by-one matrices  $A + \lambda B$ .
- `pencil_unique_eigenvalue` identifies its unique root; `pencil_charpoly_separable` proves separability.
- `collision_set_empty` proves that the set defined by simultaneous vanishing of the characteristic polynomial and derivative is empty. Its `Nat.card` is consequently an ordinary finite cardinality; no convention for infinite cardinalities is used.
- `expected_crossing_count_zero` evaluates the actual integral for every measure and every pair of matrix-valued sample functions. `conjectured_expectation_false` gives the contradiction for every probability measure.

All final theorems use only Lean’s standard logical axioms. The project contains no unproved placeholders, custom axioms, or native evaluation used as a proof. No auxiliary computation is required.

**Reproduction.** In `lean/`, run:

```
lake build
lake env lean -DwarningAsError=true Main.lean
```

Lean 4.19.0 and the exact Mathlib revision are pinned in the project. Run `tectonic main.tex` from the submission directory to rebuild this PDF. The bilingual source is preserved in `SOURCE.md`.

### References

- [1] The Last Math Competition, Conjecture 00000002343.
- [2] The Mathlib Community, Lean 4 mathematical library, revision c44e0c8ee63c.